

Machine Learning

Linear Regression- Week 4

Objectives

- Cost Function
- Intuition behind Cost Function
- Reading Assignment (Andrew Ng , CS229 Lecture notes)

Slide credits:

Andrew Ng, CS229 Lecture notes

Where to find me?

Office:

Science Hall 250

Office Hours:

**Weds and Thurs 11:00 AM - 1:00 PM EST or by
appointment**

Terminology

x = input variable or features or attributes

y = output variable/ target

m = total number of training samples

In our case, m is equal to 47 samples

We denote a single training sample as

$(x^{(i)}, y^{(i)})$

Example, (x^1, y^1) is (2104, 400)

Living area (feet ²)	Price (1000\$)
$x^1 \rightarrow 2104$	$y^1 \rightarrow 400$
$x^2 \rightarrow 1600$	$y^2 \rightarrow 330$
2400	369
1416	232
3000	540
$\vdots$	$\vdots$

Linear Regression

The equation for linear regression, therefore, can be written as:

$$f_{w,b}(x) = w * x + b$$

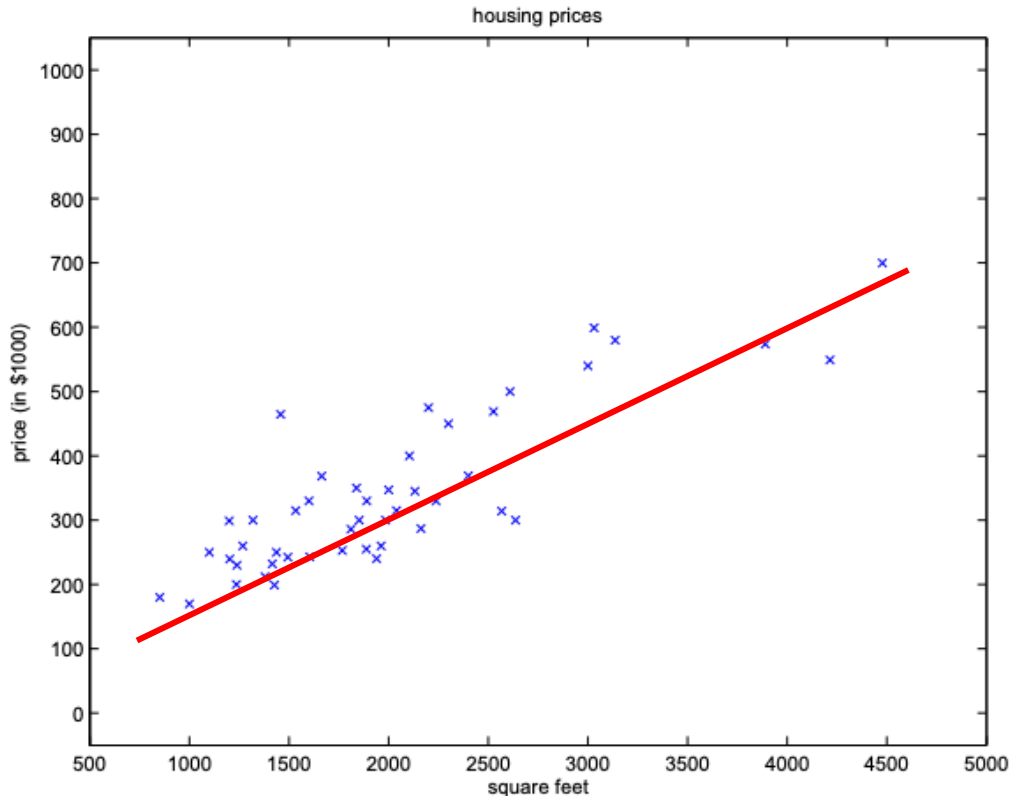
$w, b \rightarrow$ parameters/coefficients/weights and bias

$f(x)$ is the estimated value of y , in our case price(in \$1000)

We often denote $f(x)$ as $\hat{y}$

This is an example of **univariate linear regression**.

Univariate $\rightarrow$ “uni” or **one** & “variate” or **variable**



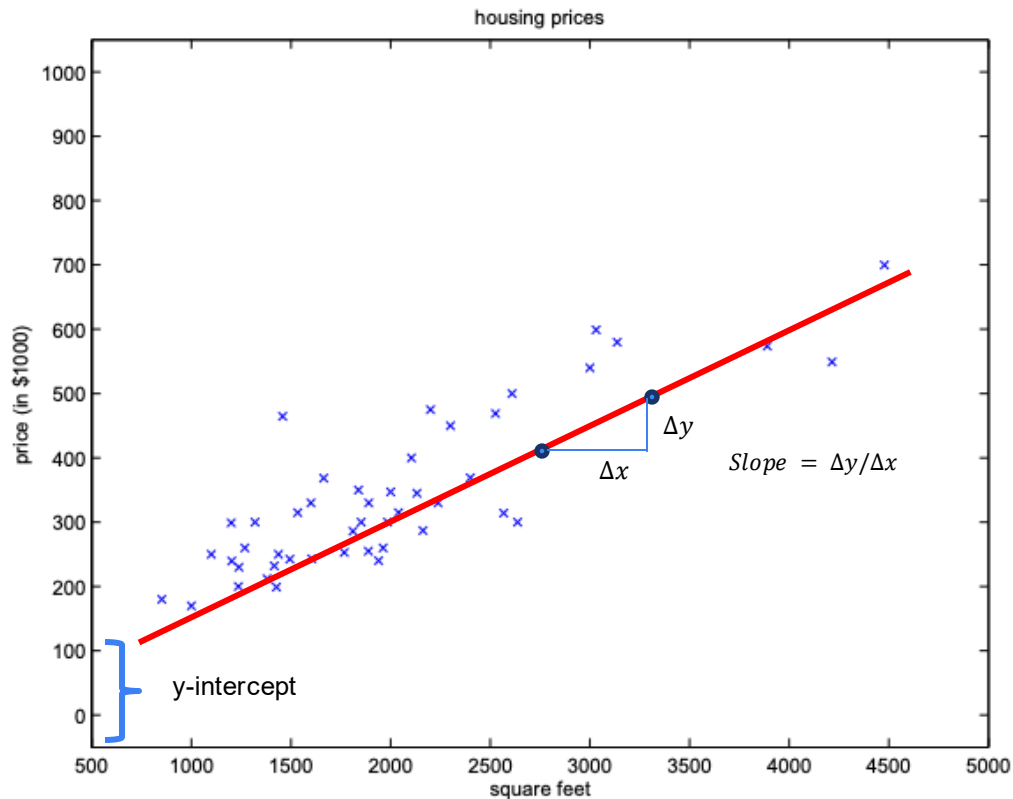
Linear Regression

The equation for linear regression,

$$f_{w,b}(x) = w * x + b$$

$w \rightarrow$ slope of the line

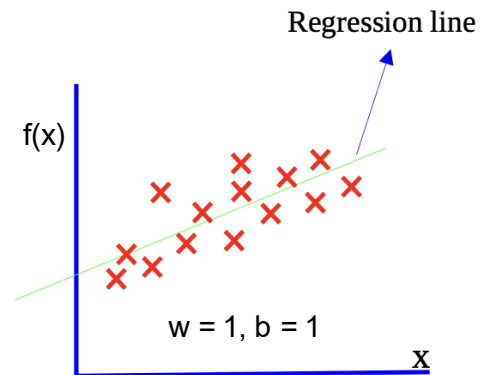
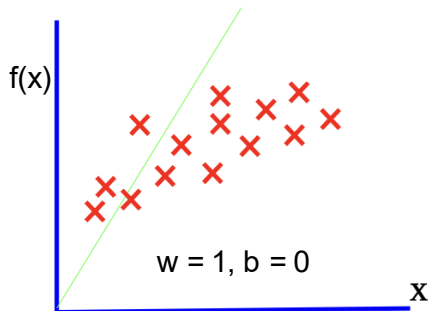
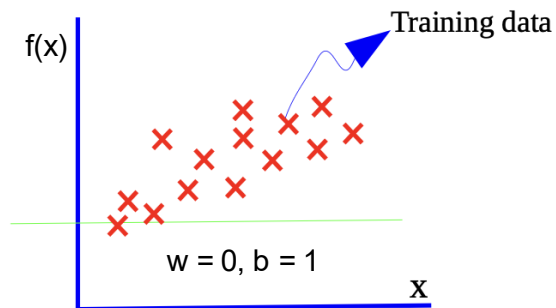
$b \rightarrow$ y-intercept



Significance of w and b

$$f_{w,b}(x) = w * x + b$$

Different values of w and b corresponds to different lines

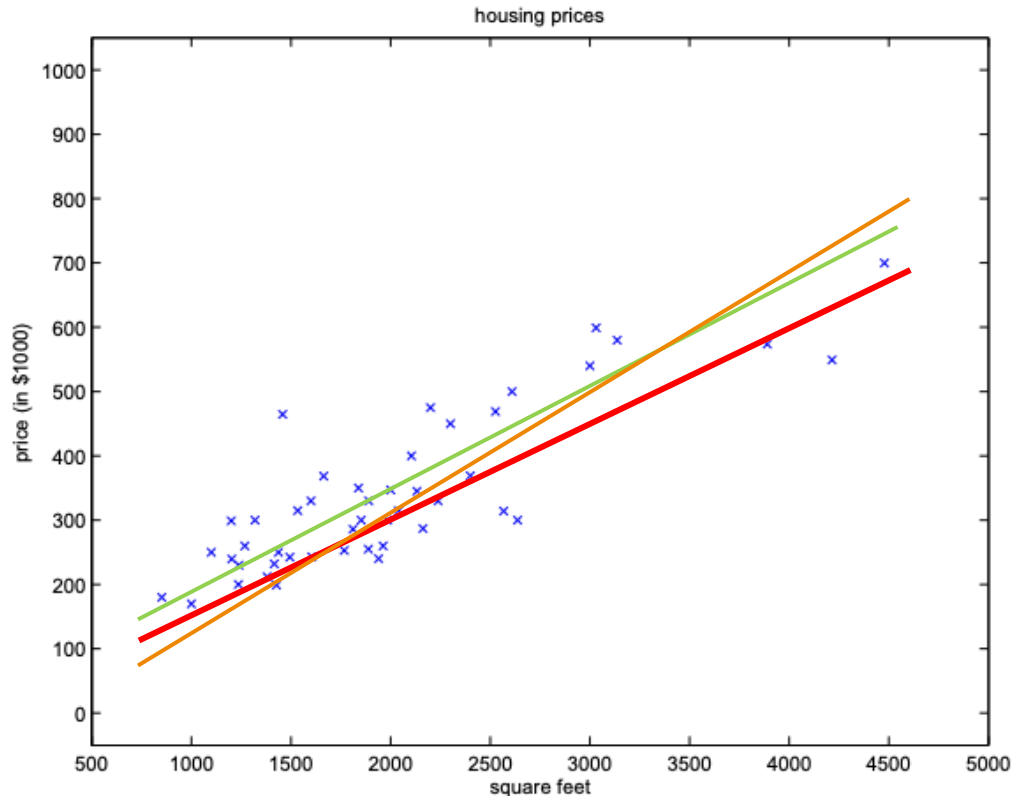


Finding the “best” fit...

For any data, we can get multiple lines or fits.

Out of these we need to select the one that passes through most of the training samples such that error between predicted values of y , i.e. $\hat{y}$ and actual value, y is minimized.

We will look at the error and cost function in next class



Cost function

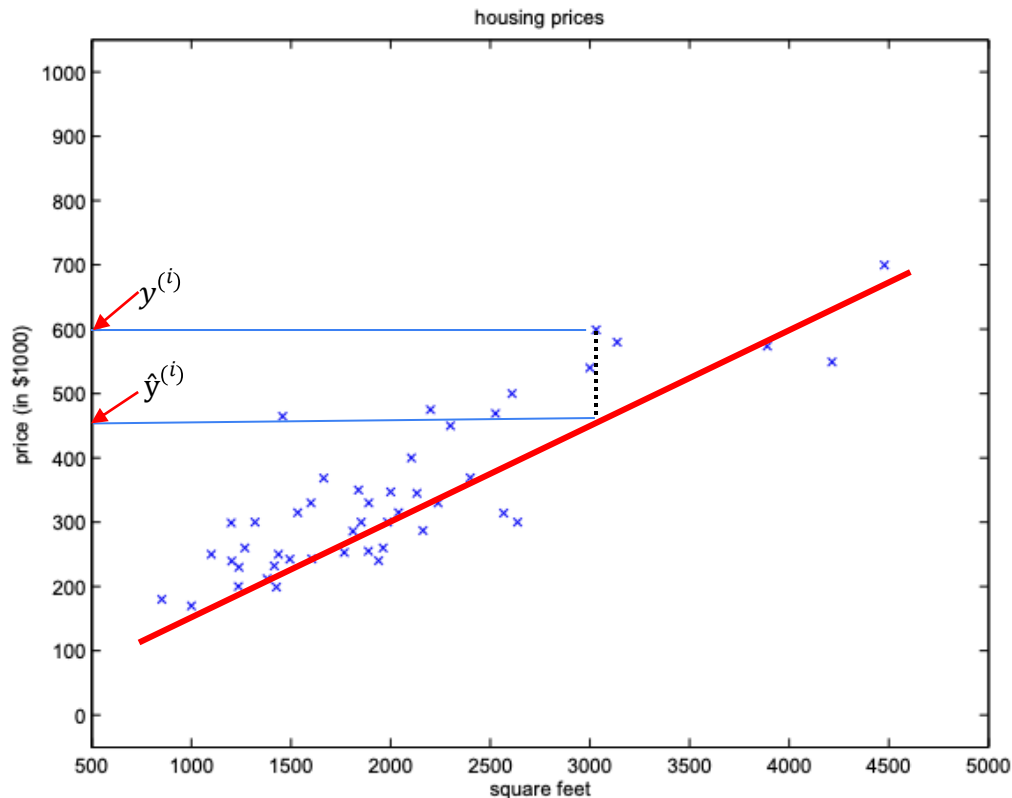
We need a way to find the values of w , b such that the linear regression line we generate roughly passes through the training samples.

The cost function helps us to find that.

For linear regression, the cost function, $J(w, b)$ usually measured as follows:

$$\frac{1}{2m} \sum_{i=1}^m (\hat{y}^{(i)} - y^{(i)})^2$$

This is often known as squared error cost function.



Cost function

For linear regression, the cost function, $J(w, b)$ usually measured as follows:

$$J(w, b) = \frac{1}{2m} \sum_{i=1}^m (\hat{y}^{(i)} - y^{(i)})^2$$

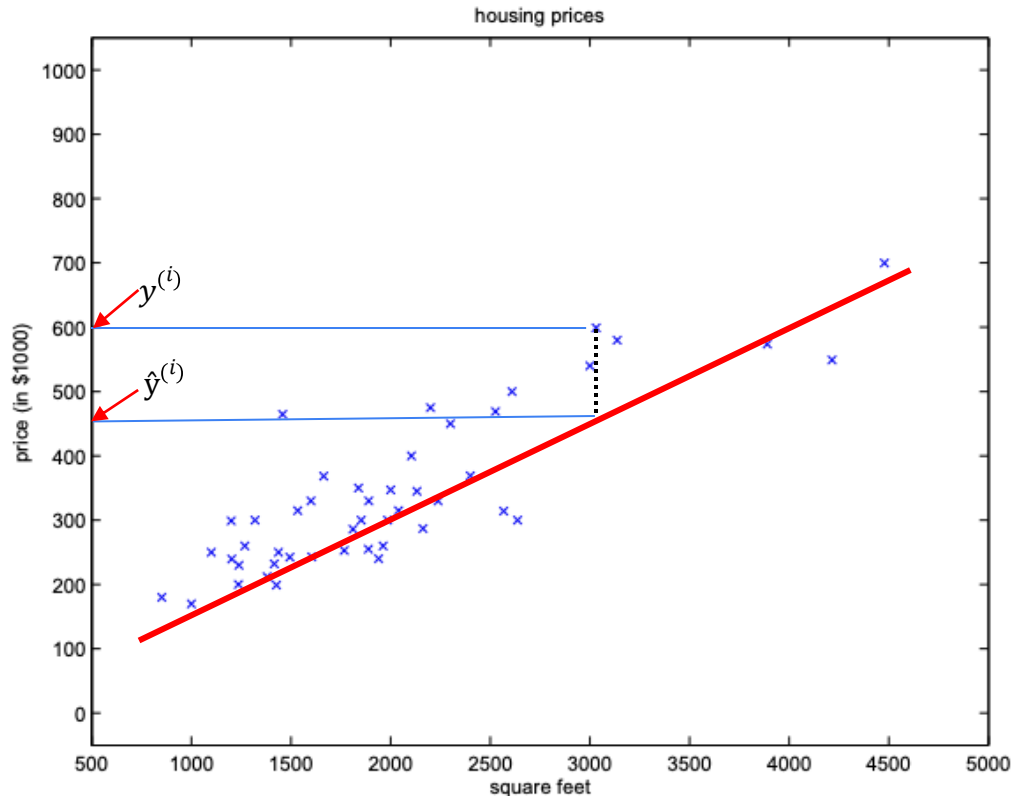
or,

$$\frac{1}{2m} \sum_{i=1}^m (f(x^{(i)}) - y^{(i)})^2$$

Substituting values for $f(x)$, we get:

$$\frac{1}{2m} \sum_{i=1}^m ((w^{(i)} x^{(i)} + b) - y^{(i)})^2$$

Our goal is to **minimize** $J(w, b)$



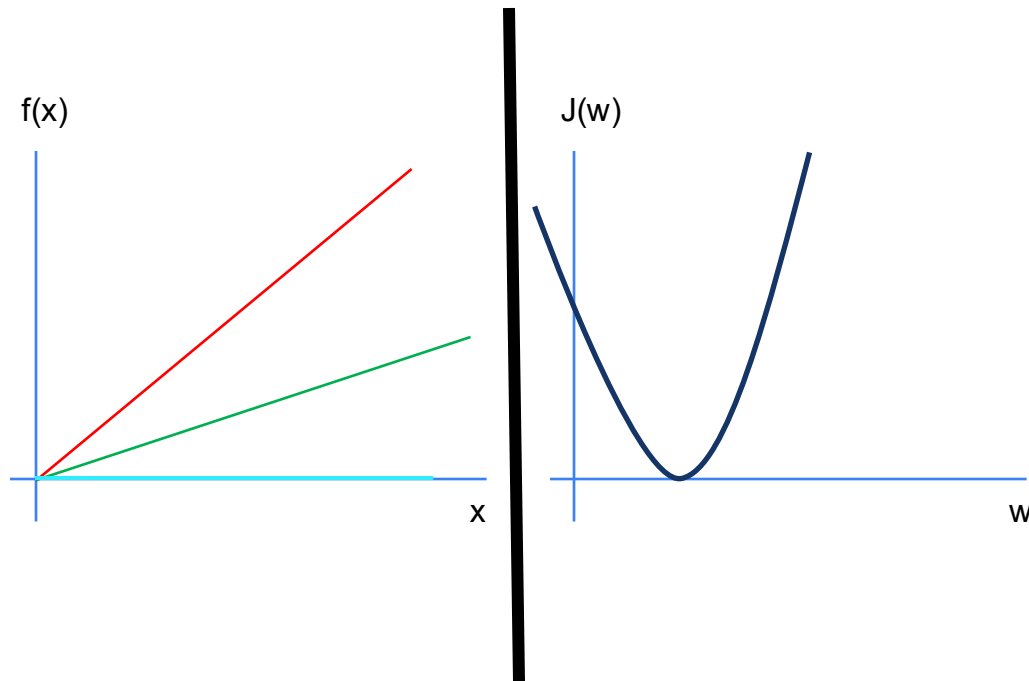
Cost function

For simplicity, we will get rid of b for now.
Therefore, let's assume $f(x) = wx$.

Substituting values for $f(x)$, we get:

$$J(w) = \frac{1}{2m} \sum_{i=1}^m (w^{(i)} x^{(i)} - y^{(i)})^2$$

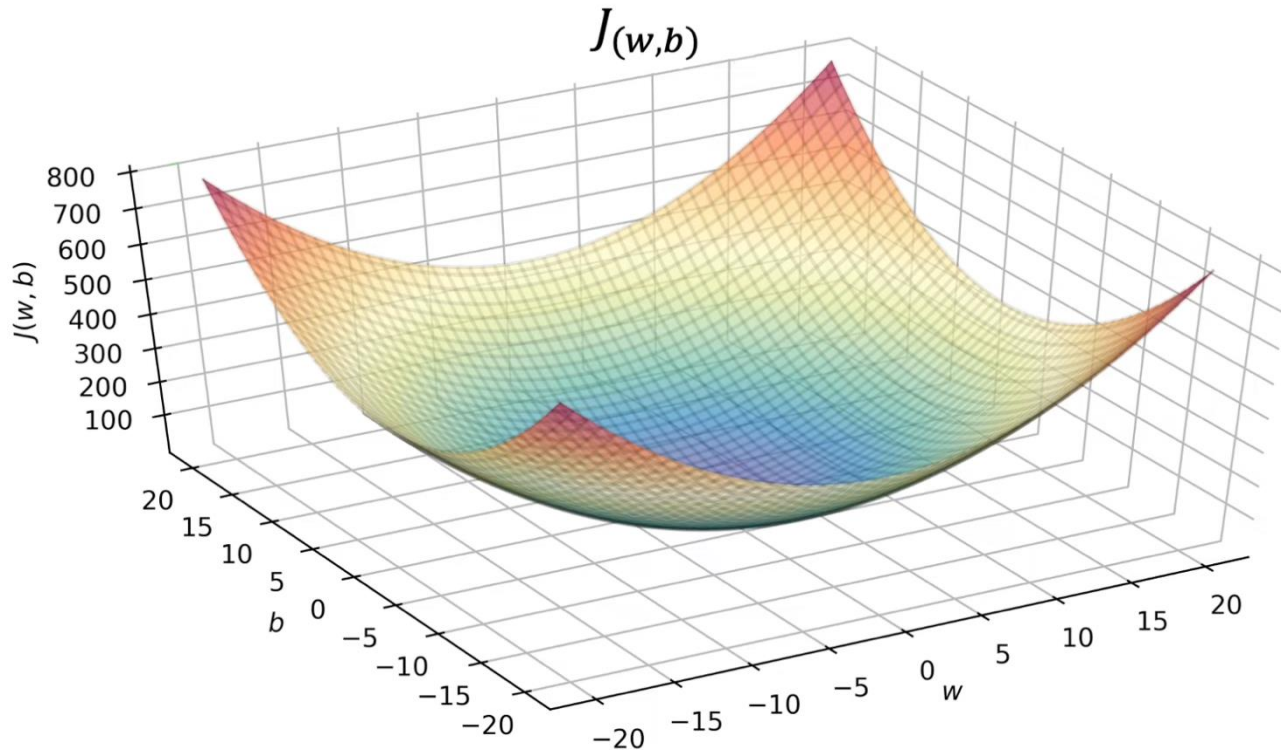
Note that $f(x)$ is a function of x whereas
cost function $J(w)$ is a function of w .



Plotting $f(x)$ and $J(w)$ for different values of x and w

Cost function

Cost function $J(w, b)$ is a function of w and b .



ANY QUESTIONS?